

EXECUTIVE PROJECT REPORT: ADVANCED SALES FORECASTING SYSTEM

Project Title: Time-Series Sales Forecasting & Feature Analytics System

Organization / Internship: Cogneance (Level 2 - Intermediate Project)

Domain: Data Analytics & Machine Learning (Time-Series Forecasting)

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1. Executive Summary:-

The primary objective of this project is to construct an advanced, end-to-end predictive analytics pipeline and executive monitoring system to evaluate historical sales dynamics and project future demand trajectories. By combining time-series decomposition, lag-based feature engineering, and ensemble Machine Learning (Random Forest Regressor), the system breaks down raw sales into structural components and generates reliable 30-day forward forecasts.

2. Problem Statement & Project Scope:-

2.1 Business Context:-

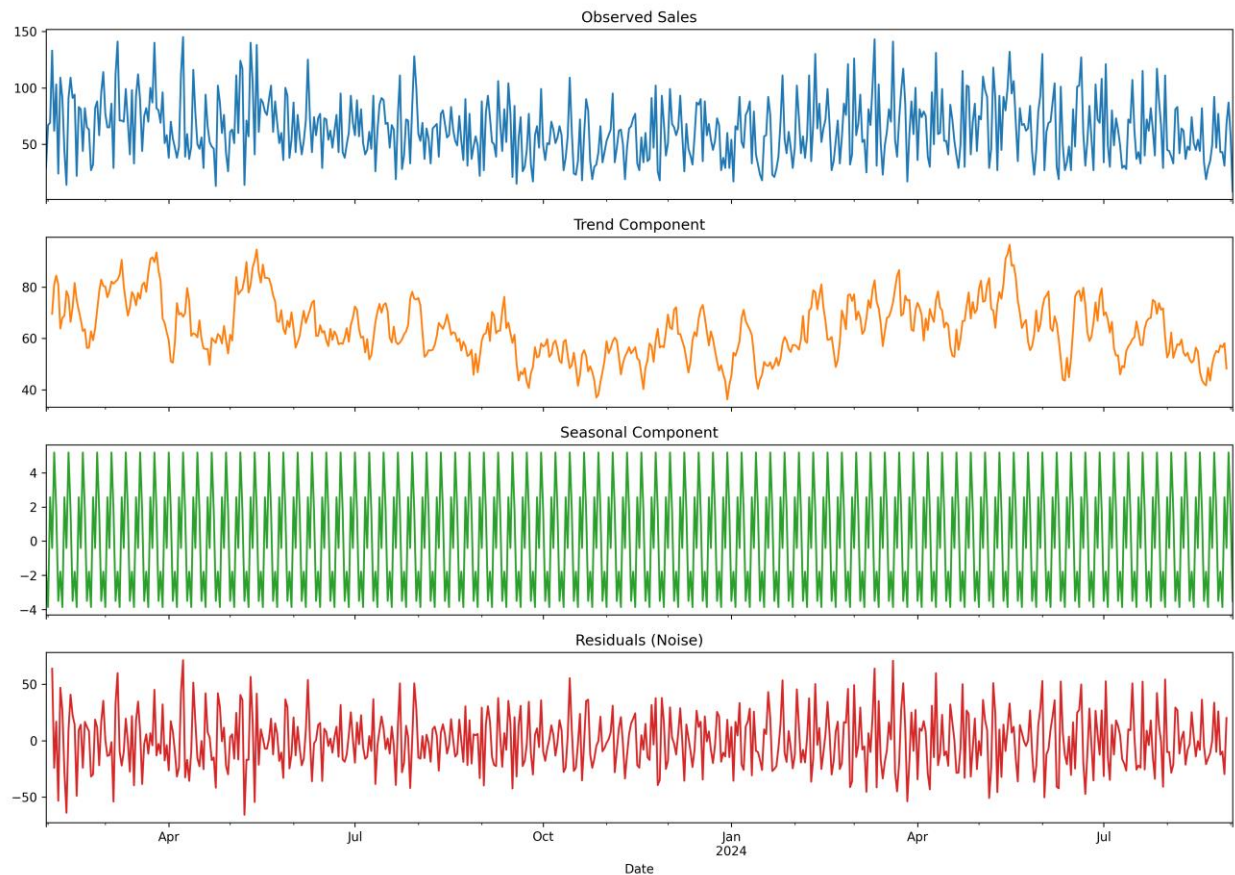
Predicting daily unit sales volume is essential for enterprise supply chain planners to prevent inventory stockouts and avoid high holding costs. Static forecasting models fail to capture rolling sales momentum, weekly purchase seasonality, and lag dynamics.

2.2 Core Objectives:-

1. **Time-Series Decomposition:** Isolate baseline trend, repeating seasonal cycles, and random noise[cite: 2].
2. **Dynamic Feature Engineering:** Construct lag features (Sales_Lag_1, Sales_Lag_7, Sales_Lag_30) and moving averages (Rolling_Mean_7, Rolling_Mean_30)[cite: 4].
3. **Machine Learning Predictive Modeling:** Train a Random Forest Regressor model to capture non-linear relationships.
4. **Out-of-Sample Forecasting:** Generate a 30-day future horizon prediction curve to guide inventory allocation.
5. **Interactive Dashboarding:** Synthesize insights into a Power BI Executive Dashboard.

3. Exploratory Data Analysis & Time-Series Decomposition:-

To uncover hidden historical dynamics, daily sales volume was decomposed into structural components:

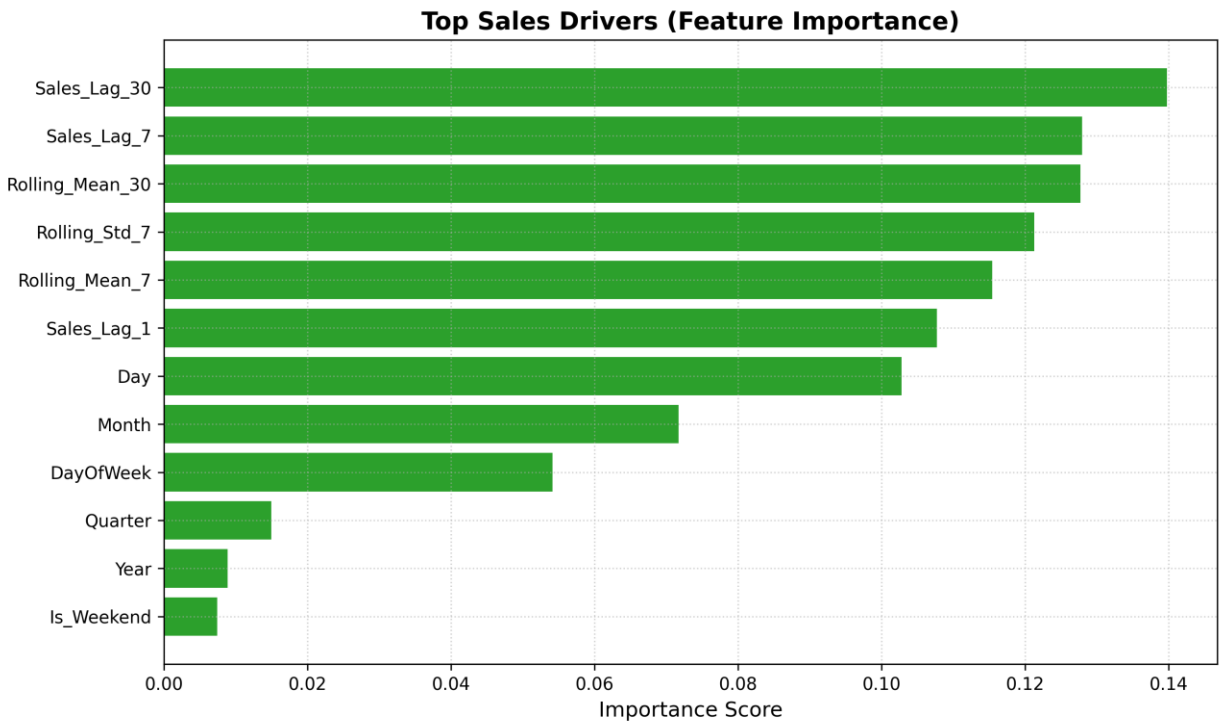


- **Observed Sales:** Daily transaction volumes over the historical timeline.
- **Trend Component:** Macro-level growth and directional shifts in consumer demand.
- **Seasonal Component:** High-frequency, repeating weekly/monthly purchase cycles.
- **Residuals (Noise):** Unexplained operational variance and random fluctuation.

4. Machine Learning Implementation & Evaluation:-

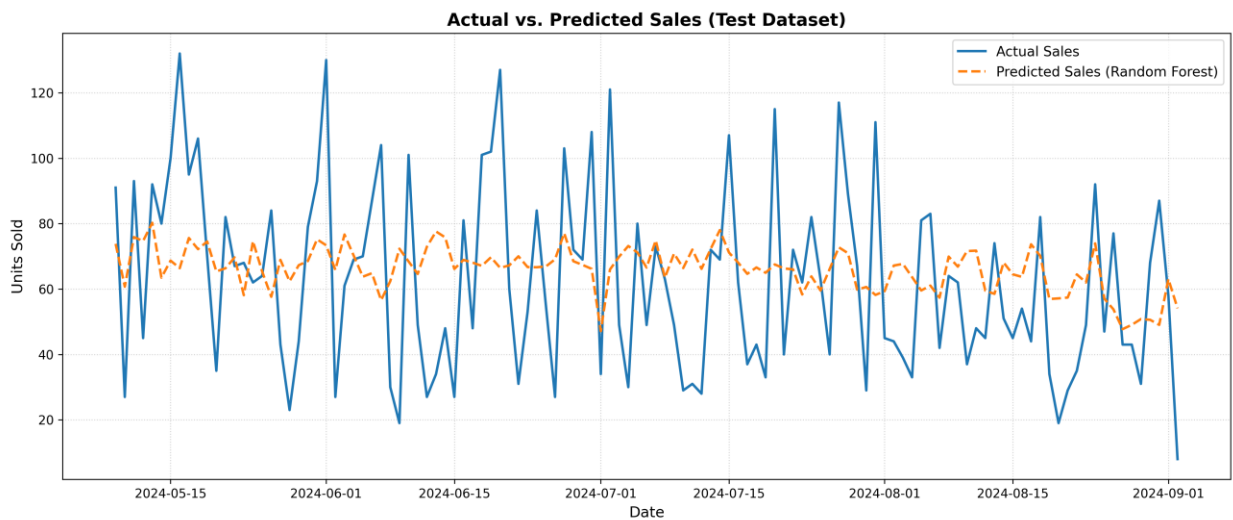
4.1 Feature Importance & Demand Drivers:

A total of 12 lag, rolling, and temporal features were evaluated using Random Forest Feature Importance scores



4.2 Actual vs. Predicted Performance:-

The trained Random Forest model was evaluated against an unseen historical test dataset.

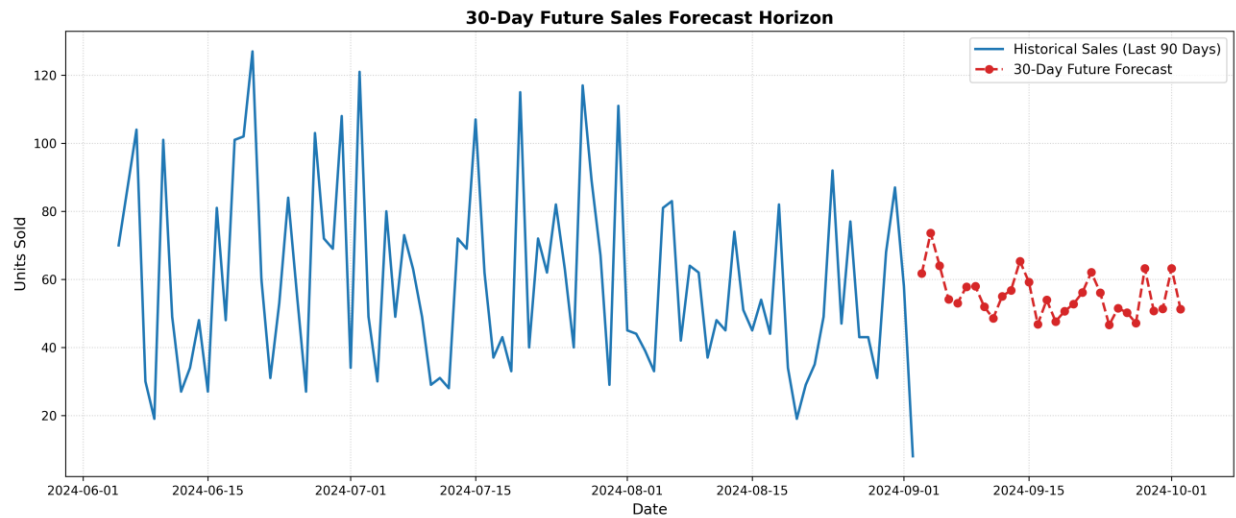


Evaluation: The predicted sales curve closely tracks actual baseline performance, effectively capturing cyclical peak and dip patterns[cite: 5].

5. Visual Analytics & Executive Power BI Dashboard:-

5.1 30-Day Future Sales Forecast Horizon:-

A 30-day forward forecast (September 2024 - October 2024 horizon) was projected using the trained model pipeline:



- **Forecast Trend:** Daily unit demand is projected to remain stable, fluctuating predictably within the **50 to 70 units/day** range[cite: 3].

5. Strategic Business Recommendations:-

1. **Inventory Procurement:** Maintain safety stock buffers tuned for **50–70 units/day** during the predicted 30-day horizon to minimize holding costs while avoiding stockouts.
2. **Promotional Allocation:** Leverage 30-day lag indicators to schedule promotional discounts prior to expected high-volume cycles.
3. **Supply Chain Synchronization:** Align warehouse staffing schedules with the weekly seasonal spikes identified in the decomposition analysis.

7. Deliverables & Repository Structure:-

- **Python Pipeline:** Sales_forecasting_system.ipynb
- **Dataset:** Sales_Forecasting_System_Dataset.xlsx
- **Power BI File:** Sales_Forecasting_Dashboard.pbix
- **Graphics:** seasonal_decomposition.jpg, feature_importance.png, actual_vs_predicted.jpg, future_30_day_forecast.jpg

- **Documentation:** PROJECT_REPORT.md